Assignment Day 3
C++ Programming

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17. Accept n numbers from users and store them in to array. Write a function to sort and display sorted array.
i/p:[1,4,32,5]
o/p:[1,4,5,32]

Program:

```
#include <iostream>
using namespace std;

// function to sort array
void sortArray(int arr[],int size){
   int temp;
   for (int i = 0; i < size; i++)
   {
       for (int j = i+1; j < size; j++)
       {
           if (arr[i] > arr[j])
           {
               temp = arr[i];
               arr[i] = arr[j];
               arr[j] = temp;
           }
       }
   }
}

// funtion to print array
void displayArray(int arr[], int size){
   for (int i = 0; i < size; i++)
   {
       cout << arr[i] << "  ";
   }
}


int main (){
```

```
    int n;
    cout << "Enter the size of array : " ;
    cin >> n;

    int arr[n];
    // to take input of the element in array
    cout << "Enter " << n << " element in array : ";
    for (int i = 0 ; i < n ; i++){
        cin >> arr[i];
    }

    // funtion call
    displayArray(arr,n);
    cout << endl;
    sortArray(arr,n);
    displayArray(arr,n);

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day3 % cd "/Users/abhiny
y3/"sortedarray
Enter the size of array : 4
Enter 4 element in array : 1 4 32 5
1  4  32  5
1  4  5  32  %
(base) abhinyaay@Abhinays-MacBook-Air day3 %
```

18. Accept n characters from user and search a character appears how many times in the array.
i/p:['a','b','c','d','a']
search 'a'
o/p: a count is 2

Program:

```
#include<iostream>
using namespace std;

// funtion to search and display the count
int searchArray(char arr[], int size, char target){
    int count = 0;
    for (int i = 0 ; i < size ; i++){
        if (arr[i] == target){
```

```
            count++;
        }
    }
    return count;
}

// funtion to diplay the array
void displayArray(char arr[], int size){
    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << "  ";
    }

}

int main(){
    int n , count;
    char ch;
    cout << "Enter the size of array : ";
    cin >> n;

    char arr[n];
    cout << "Enter  " << n << " character element in array : ";
    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    // funtion call ;
    displayArray(arr,n);
    cout << endl;
    cout << "Enter the char to search in the array : ";
    cin >> ch;
    count = searchArray(arr , n , ch);
    cout << ch << " apperared " << count << " times in the array " << endl;
    return 0;

}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day3 % cd "/Users/abhiny
y3/"searcharray
Enter the size of array : 5
Enter  5 character element in array : a b c d a
a  b  c  d  a
Enter the char to search in the array : a
a apperared 2 times in the array
(base) abhinyaay@Abhinays-MacBook-Air day3 %
```

19. Accept n numbers from user and write a function to find out average and display average.
i/p:{10,20,30}
o/p:Average: 20

Program:

```
#include<iostream>
using namespace std;

// Function to find average
int averageArray (int arr[], int size)
{
   int sum = 0;
   int avg;
   for(int i = 0 ; i < size ; i++)
   {
       sum = sum + arr[i];
   }
   avg = sum / size;
   return avg;
}

int main ()
{
   int n , avg;
   cout << "Enter the size of array : " ;
   cin >> n;

   int arr[n];
   cout << "Enter the element in array : ";
   for(int i = 0; i < n ; i++)
   {
       cin >> arr[i];
   }
   avg = averageArray(arr , n);
```

```
    cout << "Average of the array is : " << avg;


    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day3 % cd "/Users/abhiny
day3/"averageArray
Enter the size of array : 3
Enter the element in array : 10 20 30
Average of the array is : 20%
(base) abhinyaay@Abhinays-MacBook-Air day3 %
```

20. Accept n numbers from user and write a function to multiply each num-ber by 2 and display result.
i/p: [10,20,30,40]
o/p:[20,40,60,40]

Program:

```
#include<iostream>
using namespace std;

// funtion to multiply number by 2
void mulArray(int arr[], int size){
    for(int i = 0; i<size ; i++){
        arr[i] = arr[i] * 2;
    }
}

// funtion to display array
void displayArray(int arr[], int size){
    for (int i = 0 ; i < size ; i++){
        cout << arr[i] << "  ";
    }
    cout << endl;
}

int main(){
    int n;
    cout << "Enter number of elements " ;
    cin >> n;

    int arr[n];
    cout << "Enter the elements in array : ";
    for (int i = 0; i < n ; i++){
```

```
        cin >> arr[i];
    }
    // function call
    cout << "Normal array : " << endl;
    displayArray(arr,n);
    cout << "Array after Multiplying by 2 : " << endl;
    mulArray(arr,n);
    displayArray(arr,n);
    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day3 % cd "/Users/abhiny
 /day3/"multiplyArray
Enter number of elements 4
Enter the elements in array : 10 20 30 40
Normal array :
10  20  30  40
Array after Multiplying by 2 :
20  40  60  80
(base) abhinyaay@Abhinays-MacBook-Air day3 %
```

21. Accept n number from user and write functions for finding min, max, second max and second min from these numbers
i/p: [10,20,30,40,5,7,10]
o/p:min:5
second min: 7
max:40
second max:30

Program:

```
#include<iostream>
using namespace std;

// funtion to sort array
void sortArray(int arr[], int size){
    int temp;
    for (int i = 0 ; i<size ; i++){
        for (int j = i+1 ; j < size ; j++){
            if (arr[i]>arr[j]){
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
```

```
    }

}

int main(){
    int n;
    cout << "Enter the number of element : ";
    cin >> n;

    int arr[n];
    for (int i = 0; i<n ; i++){
        cin >> arr[i];
    }
    sortArray(arr,n);
    cout << "MAX value : " <<  arr[n-1] << endl;
    cout << "MIN value : " <<  arr[0] << endl;
    cout << "SECOND MAX value : " <<  arr[n-2] << endl;
    cout << "SECOND MIN value : " <<  arr[1] << endl;
    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air day3 % cd "/Users/abhiny
y3/"minmaxArray
Enter the number of element : 7
10 20 30 40 5 7 10
MAX value : 40
MIN value : 5
SECOND MAX value : 30
SECOND MIN value : 7
(base) abhinyaay@Abhinays-MacBook-Air day3 %
```